Presentation 4

• Links to presentation(s) and code(s) on GitHub

New (full data) Benefits Menu Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/BenefitsMenuOutcomes_Nov18_Marcos.ipynb

New (full data) Pre legal seniors menu code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/PreLegalSeniorsOutcomes_Nov18_Marcos.ipynb

Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/SeniorsAbandonment_18Nov_Marcos.pdf

• What did you do?

I worked on getting an understanding on call outcomes and particularly call abandonments within the pre legal seniors menu. I did this by finding all calls that went through the seniors menu at some point and then getting their entire call path including other menus. I then looked for what activity these calls ended up on and if they ended up on the seniors menu, I identified if they should be classified as an abandoned call.

• How does it help the project?

This helps give us an idea of if any single menu, verbiage, etc. is accounting for a disproportionate amount of call abandonments. If this is the case we can look for ways to improve it. At this point less than 3% of the calls that go through the seniors menu are abandoned within the menu and therefore is a positive sign overall.

• Issues faced (if any)

I did not face any issues. The only struggle I had was labeling some scenarios as abandoned or not because it was a bit complicated

• Attempts to resolve issues (if any)

N/A

• Issues resolved (if any)

N/A

• Next steps

As a team we will be calling legal aid to confirm all of our labelings when it comes to abandonment to ensure accuracy of the numbers we currently have

• References (Mention if you built up on someone else's work)

Presentation 3

• Links to presentation(s) and code(s) on GitHub

Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/BenefitsMenuAbandoned_4Nov_Marcos.ipynb

Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/BenefitsAbandonment_5Nov_Marcos.pdf

• What did you do?

I worked on getting an understanding on call outcomes and particularly call abandonments within the benefits menu. I did this by finding all calls that went through the benefits menu at some point and then getting their entire call path including other menus. I then looked for what activity these calls ended up on and if they ended up on the benefits menu, I identified if they should be classified as an abandoned call.

• How does it help the project?
This helps give us an idea of if any single menu, verbiage, etc. is accounting for a disproportionate amount of call abandonments. If this is the case we can look for ways to improve it. At this point less than 5% of the calls that go through the benefits menu are abandoned within the menu and therefore is a positive sign overall that it is not discouraging callers and making them hang up.

• Issues faced (if any)
An issue that the team and i faced was that we were all getting different numbers of data when we loaded the CAR data into our devices. This was a problem because we all are focusing on a specific menu, but if we are working off of different numbers from the start, the results will be inconsistent and inaccurate

• Attempts to resolve issues (if any)
We redownloaded all of the data files and ensured to all begin working from March 16th data in order to be consistent.

• Issues resolved (if any)
Our inconsistent data issue was fixed. By the end, we all had 90,940 unique calls since March 16th, so we are sure we are working with the same numbers

• Next steps
As a team we hope to continue looking at some of the other numbers to eventually get the entire picture of call abandonment throughout the entire call flow

• References (Mention if you built up on someone else's work)

Presentation 2
• Links to presentation(s) and code(s) on GitHub
Code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Chatbot%20analysis/ChatbotAnalysisCode_20Oct_Marcos.ipynb
Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Chatbot%20analysis/ChatbotAnalysis_20Oct_Marcos.pptx

• What did you do?
I completed tasks relating to analysis of the chatbot. I redid the analysis that I did last time with some tweaks. I analyzed how termination reasons look different for calls that reached the chatbot vs some that did not (I needed to redo this because I got a large number of N/A's as termination reasons in my first analysis). I also analyzed the duration of calls before, during, and after chatbot usage using a boxplot, which gave me a better idea of the distribution of duration within each period. Finally I looked at what people spent their time on within the family menu and how that looked different in the different periods i described above.

• How does it help the project?
It helps the project by getting a better understanding of the chatbot's impact on the customer experience and ultimately will help inform the decision to purchase or not purchase it moving forward
• Issues faced (if any)
The CAR data does not have a specific column specifying how long someone was doing an activity for so seeing what people spent time on was more difficult.

• Attempts to resolve issues (if any)

In order to solve this, I tracked the contact session ID and tracked the time between when someone started doing an activity and when it was logged that they started doing something else.

• Issues resolved (if any)

• Next steps
In the termination reason analysis I will limit the control group to calls that reached a human agent or compare termination reasons during the broad chatbot period vs before/after the chatbot period because at this point we might be comparing apples to oranges. The reason why agent left termination reason for the chatbot reached calls is so much higher than those that didn't may be because most calls just don't reach an agent at all

• References (Mention if you built up on someone else's work)

Presentation 1
• Links to presentation(s) and code(s) on GitHub
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/tree/main/Code/Chatbot%20analysis
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/tree/main/Presentations/Chatbot%20analysis

• What did you do?
I did the chatbot analysis part of the presentation. More specifically I looked to do an analysis on the termination reasons when chatbots were and were not used post-March (when the termination column was added) and a duration of calls analysis during the time period that the chatbot was in use vs. when it was not. The termination reason analysis was meant to provide insight on if the call was successful in helping the client or not, and the call duration analysis was meant to analyze efficiency with and without the chatbot

• How does it help the project?
This helps the project because it helps get us closer to answering the question of the effectiveness of the chatbot. This is an important business decision that Cynthia wants to make. Is it worth purchasing this chatbot permanently? This helps answer that question. The insight that we got from the analysis is that when we remove calls with an empty termination reason, it seems that chatbot calls are more efficient because an agent ended the call a higher percentage of the time, which we are assuming means that the issue was more likely to have been resolved. We also saw a much higher duration from the calls that had encountered the chatbot, which can be potentially inefficientness, but more analysis needs to be done to determine the reason for this.

• Issues faced (if any)
There were a lot of missing values for termination reasons amongst the contact session ID's. This can possibly impact the proportion I get when I removed these N/A's and lead to inaccurate representations of the real data and as a result, incorrect conclusions.

• Attempts to resolve issues (if any)
I tried finding out why these calls with N/As, if there was something that was different about those calls, but I couldn't find anything.

• Issues resolved (if any)

• Next steps
I will try to see if there is another way that I can possibly think about measuring outcome effectiveness. I will also shorten my termination analysis specifically to august inside of going back to March to see if I can get values for all calls during the month and compare this to the proportion that I currently have. I also would like to dig into why the chatbot calls might have had such a long duration.

• References (Mention if you built up on someone else's work)